Michael E. Rockman, M.D., Ph.D.

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EDUCATION

Degree	Institution	Dates Attended
M.D.	Temple University	2015 – 2022
Ph.D. *Biophysics*	Temple University	2015 – 2020
B.S. *Biochemistry (Mathematics Minor)*	Lafayette College	2011 – 2015

Postgraduate Training	Institution	Dates Attended
Cardiovascular Disease Fellowship	University of Wisconsin	2024 -
Internal Medicine Residency	University of Wisconsin	2022 - 2024

RESEARCH

University of Wisconsin School of Medicine and Public Health, Madison WI **July 2026 -**
Supervisor: Gail Robertson, PhD
Summary: Postdoctoral research investigating hERG channel gating mechanisms and assembly using conformation-selective antibody fragments (ScFv), integrating electrophysiology and kinetic modeling to define structure-function relationships with relevance to arrhythmia mechanisms and sudden cardiac death.

Temple University School of Medicine, Philadelphia PA **January 2016 – April 2020**
Supervisor: Brad S. Rothberg, PhD
Dissertation: Mechanisms of pharmacological activation of BK channels.
Summary: Characterized BK channel modulation by small-molecule drugs and disease-associated mutations primarily using patch-clamp electrophysiology. Integrated macroscopic currents and single-channel recordings to generate kinetic models quantifying altered channel gating.

Lafayette College, Easton PA **August 2012 – May 2015**
Supervisor: Justin K. Hines, PhD
Thesis: Mechanism of action of the J-protein Swa2 in the propagation of the yeast prion [*URE3]*
Summary: Performed plasmid-shuffling genetic techniques and tetracycline-repressible expression studies to investigate Swa2 function in yeast prion [*URE3*] propagation.

Duke University Medical Center, Durham NC **Summer 2009 and 2010**
Howard Hughes Medical Institute
Supervisor: Robert J. Lefkowitz, MD
Summary: Used a co-immunoprecipitation based assay to characterize conformation-selective synthetic antibody fragments (Fabs) that recognize and stabilize distinct β_2-adrenergic receptor states induced by biased ligands. In parallel, investigated β-arrestin-mediated signaling in mouse development through histological analysis of sectioned tissue samples.

PEER-REVIEWED MANUSCRIPTS

(#) Corresponding Author

8. [#] **Rockman ME**, Almothafer Z, Doucette R, Robbins DJ, Scolarici M, Liu M, Pepperell CS, Matkovic E, Kenik J. A case of travel associated Tick-Borne Relapsing Fever in Wisconsin. WMJ 2025. May 10

7. Yeung HM, Ifrah A, **Rockman ME**. Quantitative Analysis of Characteristics Associated with Patient-Directed Discharges, Representations, and Readmissions: A Safety-Net Hospital Experience. J Gen Intern Med. 2023 Dec 19. doi: 10.1007/s11606-023-08563-z

6. Pepin ME, Kamal Y, Reisman B, **Rockman ME**, and Waller J. Making the Match and Breaking it: Values, Perceptions, and Obstacles of Trainees applying into Physician-Scientist Training Programs. BMC Medical Education. 2023 Oct 12. doi: 10.1186/s12909-023-04736-w

5. Vouga AG, **Rockman ME**, Yan J, Jacobson MA, Rothberg BS. State-dependent inhibition of BK channels by the opioid agonist loperamide**.** J Gen Physiol. 2021 Sept 6.doi: 10.1085/jgp.202012834

4. **Rockman ME**, Vouga AG, Rothberg BS. Molecular mechanism of BK channel activation by the smooth muscle relaxant NS11021. J Gen Physiol. 2020 Jun 1. doi: 10.1085/jgp.201912506

3. Butler MW, Bociulis SC, Little AR, Minnick JA, Ritter NJ, **Rockman ME**, Rossi ML, Hines JK. Quantifying biliverdin in liver and spleen samples from multiple avian species. The Auk. 2016 Oct 12. doi: 10.1642/AUK-16-73.

2. Tomar D, Dong Z, Shanmughapriya S, Koch DA, Thomas T, Hoffman NE, Timbalia SA, Goldman SJ, Breves SL, Corbally DP, Nemani N, Fairweather JP, Cutri AR, Zhang X, Song J, Huang J, Jana F, Huang J, Barrero C, Rabinowitz JE, Luongo TS, Schumacher SM, **Rockman ME**, Dietrich A, Merali S, Caplan J, Stathopulos P, Ahima RS, Cheung JY, Houser SR, Koch WJ, Patel V, Gohil VM, Elrod JW, Rajan S, Madesh M. MCUR1 is a Scaffold Factor for the MCU Complex Function and Promotes Mitochondrial Bioenergetics. Cell Reports. 2016 May 24. doi*:* 10.1016/j.celrep.2016.04.050*.*

1. Troisi EM**, Rockman ME**, Nguyen PP, Oliver EE, and Hines JK. Swa2, the yeast homolog of mammalian auxilin, is specifically required for the propagation of the prion variant [URE3-1]. Mol Microbiol. 2015 Jun 1. doi: 10.1111/mmi.13076

MANUSCRIPTS UNDER REVIEW

2. **Rockman ME.** Closing the Certification Gap in Physician-Scientist Research Pathways

1. **Rockman ME,** Sandhu H, Kaniper S, Paynton B, Patel RK, Gulino SP, Gerhard GS, Cooper JM. A novel DSG2 genetic variant associated with a familial sudden death arrhythmia syndrome.

OTHER PUBLICATIONS

1. **Rockman, Michael E**. "A Doctor-in-training Learns Bedside Manners, the Hard Way." The Philadelphia Inquirer 16 Oct. 2016

ABSTRACTS

*(*) Oral presentation*

18. Li C, **Rockman M**, Kelly A (2026, September) Small Device, Big Decision: Leadless Pacing in a High-Risk Patient. *ACP Wisconsin Annual Scientific Meeting. Wisconsin Dells, WI*
17. Liu M, **Rockman ME**, Almothafer Z, Doucette R, Robbins DJ, Scolarici M, Pepperell CS, Matkovic E, Kenik J. (2025, April) An Unexpected Case of Non-Endemic Tick-Borne *Relapsing Fever. Society of Hospital Medicine Annual Meeting. Las Vegas, NV*
16. Vouga AG, **Rockman ME,** Flores L, Blazquez J, Brauchi SE, Rothberg BS. (2024, Feb) Structure and Function of an ATP-sensing channel from mycobacteria. *Biophysical Journal* 123, no. 3: 532a. *Biophysical Society Annual Meeting. Philadelphia, PA*
15. Pepin ME, **Rockman ME**, Kamal Y, Reisman B, and Waller J. (2023, April) Making "The Match" or Breaking it? Applying into Physician-Scientist Training Programs (PSTPs). *Joint Meeting AAP/ASCI/APSA. Chicago, IL*
14. Ifrah A, Pearigen A, **Rockman M**, Bittar Y, Yeung H (2022, October) I want to leave: A retrospective study on the factors associated with leaving AMA, representation and readmission. *American College of Physicians Pennsylvania Chapter. Philadelphia, PA*
13. * Ifrah A, Pearigen A, **Rockman M**, Bittar Y, Yeung H (2022, June) Risk factors and general trends associated with discharges against medical advice, re-presentation, and readmission. *Sol Sherry Research Symposium. Philadelphia, PA*
12. **Rockman ME**, Vouga AG, Rothberg BS (2020, February) Molecular mechanism of BK channel activation by the smooth muscle relaxant NS11021. *Biophysical Journal* 118, no. 3: 109a. *Biophysical Society Annual Meeting. San Diego, CA*
11. *** Rockman ME**, Vouga AG, Rothberg BS (2019, September) Mechanism of NS11021 activation of BK Channels. *Society of General Physiologists 73rd Annual Symposium. Valparaíso, Chile*
10. Vouga AG, **Rockman ME**, Jacobson MA, Rothberg BS. (2019, March) Mechanism of BK Channel inhibition by the opioid agonist Loperamide. *Biophysical Journal* 116, no. 3: 151. *Biophysical Society Annual Meeting. Baltimore, MD*
9. Vouga AG, Matychak KK, **Rockman ME**, Flores L, Brauchi S, and Rothberg BS. (2019, March) Multiple Nucleotide-dependant Conformations of a Mycobacterial RCK Domain. *Biophysical Journal* 116, no. 3: 39. *Biophysical Society Annual Meeting. Baltimore, MD*
8. Vouga AG, **Rockman ME**, Matychak KK, Piskorski J, Jacobson MA, Rothberg BS. (2018, July) Inhibition of BK channels by loperamide. *Gordon Research Conference - Ion Channels. South Hadley, MA*
7. Vouga AG, Matychak KK, **Rockman ME**, Garcia S, Brauchi S, and Rothberg BS. (2018, February) Crystal Structure of a Mycobacterial RCK Domain. *Biophysical Journal* 114, no. 3 (2018): 127a. *Biophysical Society Annual Meeting. San Francisco, CA*
6. Vouga AG, Hendron H, Smith F, **Rockman M**, Fouch M, Rothberg BS. (2016, July) Role of an intersubunit Ca^{2+} bridge in structure and function of BK channels. *Gordon Research Conference - Ion Channels. South Hadley, MA*
5. Hines JK, Troisi EM, **Rockman ME**, Oliver EE. (2015, April) Specificity of the Yeast Auxilin Homolog Swa2 in [URE3] Prion Propagation Is Solely Determined by the TPR Domain (713.5) *The FASEB Journal* 29 (1 Supplement), 713.5. *American Society for Biochemistry and Molecular Biology Annual Meeting. Boston, MA*

4. **Rockman ME**, Troisi EM, Oliver EE, Hines, JK. (2014, December). Using Tetracycline-Repression to Determine the Role of the J-protein Swa2 in [URE3] Prion Propagation. In *Molecular Biology of the Cell* (Vol. 24) *American Society for Cell Biology Annual Meeting. Philadelphia, PA*
3. Oliver EE, **Rockman ME**, Hines JK. (2014, December). Structural homology analyses of the tetratricopeptide repeat region of the yeast auxlin homolog Swa2 reveal considerable structural conservation with human HOP1. In *Molecular Biology of the Cell* (Vol. 25*) American Society for Cell Biology Annual Meeting. Philadelphia, PA*
2. **Rockman M**, Troisi E, Hines J. (2014, April). Ectopic expression studies of the J-protein Swa2 and its potential role in yeast prion propagation (567.2). *The FASEB Journal*, *28*(1 Supplement), 567-2. *American Society for Biochemistry and Molecular Biology Annual Meeting. San Diego, CA*
1. Troisi E, **Rockman M**, Hines JK. (2013, January). Investigating the role of the J-protein Swa2 in the propagation of the yeast prion [URE3]. In *Molecular Biology of the Cell* (Vol. 24) Amer Soc Cell Biology. *American Society for Cell Biology Annual Meeting. New Orleans, LA*

RELATED EXPERIENCE

Society of General Physiologists (January 2024 –)
Postdoc councilor of SGP (2024, 2025). Remain member of communications committee

American Physician Scientist Association (August 2016 – January 2020)
President of Temple Chapter of APSA (2019). Organized speakers, facilitated medical student research on campus, and advocated for more research in medicine.

Narrative Writing (June 2016 – May 2022*)*
Work with director of narrative medicine through individual communication and organized workshops to write about the powerful experiences in becoming a physician.

General Chemistry Lab Teacher Assistant (August 2013 – December 2014)
Oversaw, instructed, and introduced students to General Chemistry lab techniques.

American Chemical Society (August 2012 – May 2015)
Member of the Lafayette chapter of ACS. Served one year on executive board.

AWARDS

Mathilde and Louis Soloff Cardiology Prize (2022)
Awarded to the outstanding graduating medical student who has demonstrated superior knowledge, skill, and interest in the field of Cardiology.

Travel Award (2019)
Awarded Temple University Travel Award for travel to the Society of General Physiologists Annual Meeting in Valparaíso, Chile

Dr. David Schwimmer ’35 Prize (2015)
Awarded to the premedical student at Lafayette College who best represents the humanitarian, cultural, and scientific qualities required of the true physician.

Carl J. Staska Prize (2015)
Awarded to student who demonstrates highest proficiency in chemical laboratory skills

Lafayette College Deans List (All eight semesters)
Awarded to students who maintain a GPA above 3.60

Barry M. Goldwater Scholarship Nominee (Fall 2012)
Received competitive school nomination for Goldwater Scholarship as college sophomore.

PROFESSIONAL MEMBERSHIP

2024 -	American College of Cardiology
2022 -	Heart Rhythm Society
2019 -	Society of General Physiologists
2019 -	Biophysical Society
2015 -	American Physician Scientists Association

FUNDING

Physician-Scientist Training Program Research Stipend (2022 -)
Department of Medicine annual stipend for research and two conferences annually.

MD/PhD Fellowship (2015- 2022)
Awarded fellowship covering tuition and a stipend throughout completion of MD/PhD.

EXCEL Research Scholar (2012-2015)
Awarded school-wide scholarship providing funding for students for on-campus research.

Summer Research Experience Grant (R25 PA-13-341) (2009)
Awarded competitive grant supplement from the NIH intended to give aspiring scientists exposure to basic science research. (Reference Number: HL070631)

CERTIFICATIONS & SKILLS

Board Certifications: Internal Medicine (ABIM)

Certifications: Basic Life Support, Advanced Cardiovascular Life Support

Clinical Skills and Procedures: Procedural: Central venous catheter placement in internal jugular and femoral veins, transvenous pacemaker placement, Swan-Ganz pulmonary artery catheter placement, radial and femoral arterial access. Imaging: Transthoracic echocardiography (TTE) acquisition and interpretation, transesophageal echocardiography (TEE) probe placement, image acquisition and interpretation, ECG interpretation, stress ECG, nuclear stress testing and stress echo interpretation.

Computer: Microsoft Office, Adobe Photoshop, Adobe Illustrator, Igor Pro, Clampfit, Axon pCLAMP, PyMOL, BioRender

Lab: Patch-clamp electrophysiology**,** kinetic modeling of ion channels, single-channel recordings, crystal screening for protein crystallization, setting and optimizing protein crystal trays, x-ray crystallography, protein purification, quantitative fast protein liquid chromatography, FLIPR high-throughput drug screen, qPCR, site directed mutagenesis, preparative and analytical gel electrophoresis, mammalian/insect/yeast/bacterial cell culture and aseptic technique, yeast and bacterial transformations, SDS PAGE, western blotting, co-immunoprecipitation assays, restriction digestions, synthetic media preparation, paraffin sectioning, OCT sectioning, H&E staining, fluorescence microscopy.